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THE RACE TO SPACE

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You're going to be thoroughly spoilt for choices when planning a holiday a century from now. You will still be able to go to the Grand Canyon or the Great Barrier Reef, but you will have to weigh those options against taking a weekend flight to the Moon and back. On the Moon you could take a tour of historic landing sites, check out artefacts, visit planned outposts, take a dune buggy ride across the dusty plains, and click a selfie with the Earth in the background (sadly, yes, selfies will still be around). You could also pop in to visit the astronauts on the space station on your way back, before finally riding a rollercoaster-like descent vehicle back into the atmosphere for kicks.

Today, we're already taking baby steps towards realising this seemingly fantastic vision. Space is finally opening up to someone other than governments, and that's a good thing. Private companies need to make a profit in the end, and that means a rush to improve technologies and reduce costs. Unlike government aided space programs, the private industry needs to see a clear path to profitability, and every dollar spent is duly accounted for by shareholders.

Rocket engines today are almost exactly what they were during the Cold War. And the monopoly of a few entities manning the gateway to space has to be broken. ULA, America's leading rocket launch service provider, charges over \$225 million per launch, as stated on their website. A more recent launch, which we shall look at in a bit, cost just one-fifth of this, at \$54 million. It's not just private companies aiding in this reduction of prices, it's also

How is today's space race shaping the present and the future of space exploration?

a more global push for space by more governments. India, as you all remember, recently led the cheapest successful mission to Mars, ever.

However, it's a bunch of pioneers with keen business acumen that will really make space more accessible to everyone.

What is Space?

Talking about exploring space conjures up images of people in rockets and spaceships, flying off to visit planets. Very Star



Keep it classy Mr. Bezos!

Trek- and Star Wars-ish. However, where does space really start? Nowhere specific, to be honest because the atmosphere doesn't abruptly stop existing, it just gradually thins out into vacuum. So we can't really use the amount of molecules per given volume, but we can use temperature. As you go higher, temperature goes down. Anyone who's been in an aeroplane knows that it's ridiculously cold outside, but pretty warm inside thanks to the air pressure. But what temperature should

we choose? So how do we go about deciding where space starts? The Kármán line, named after Theodore von Kármán, who calculated that if you were in an aircraft, and kept going higher, at one point you'd have to be going exactly at orbital velocity to get the required lift to go any higher. Orbital velocity is the point at which the attraction of gravity and inertia of a body to move out in a straight line instead of following the Earth's curvature balance out, and you achieve a stable orbit. This point is about 100 km above the surface of the Earth. So if you're flying below the Kármán line you're a mere pilot, and above, you're an astronaut (or cosmonaut).

Space tourism

Obviously, when people are talking about space tourism, they're talking about taking people above the Kármán line, and obviously, bringing them back.

Anyone who's seen videos of the Challenger disaster and then the later Columbia one (1986 and 2003 respectively), know that space is not a game. It's dangerous business. It's hard enough to go to space, but coming back to Earth is just as risky.

While Challenger broke up during take off, Columbia broke up during re-entry, underscoring the very real and dangerous risks of spaceflight. A lot of us in India remember the Columbia disaster, not only because it is more recent, but because amongst the people killed was Kalpana Chawla, who was the first woman of Indian origin in space.

Any craft that enters the atmosphere from space has to penetrate it at precisely the right angle. Otherwise, it might